

Market Cluster Dynamics — S&P 500 Prototype (2026-09-28, rev. 3)

The problem with rev. 1

The stationary-drift model (rev. 1, 2026-09-27) reproduced calm-market distributions (RMSE 0.045) but could not generate the observed spikes: March 2021 saw 55% of stocks pile into the top momentum bin (>60% twelve-month return); late 2008 saw 98% pile into the bottom bin (<0%). A monthly Markov transition matrix mean-reverts too fast to build or hold such concentrations. Herding (bilinear flux toward above-average-momentum bins) was a weak amplifier at best — it can only redistribute mass the drift already moves.

New reaction kernels (the mechanism first)

Four kernels. Each has a physical mechanism, not just a fitted parameter.

1. Market advection (the spike generator)

Mechanism. When the market's trailing twelve-month return shifts by M in a month, every stock's momentum shifts by $\sim M$ in return space — a rigid translation of the whole distribution along the momentum axis. This happens mechanically at window rollover: the month entering the 12-month window replaces the month dropping out. The driver is

$$M(t) = r(t) - r(t-12)$$

the *synchronized change in twelve-month momentum*, measured cross-sectionally (equal-weighted over all index stocks, M_{cs}). March 2021: $M_{cs} = +0.256$ (the -12% March-2020 crash month exited the window as the +15% March-2021 month entered). October 2008: $M_{cs} = -0.256$.

Implementation. Donor-cell advection within each volatility lane: a shift M moves fraction

$$f = \kappa |M| / w_{\text{eff}}$$

of each bin to the next, $\kappa = 1.0$ (the *mechanical* value, not fitted), $w_{\text{eff}} = 0.15$ assumed (typical momentum-bin width). Boundaries reflect: the extreme bins ($<0\%$, $>60\%$) are unbounded, so mass that reaches them accumulates — this is what generates spikes. M_{cs} is **measured** from Yahoo Finance daily closes.

Timing note. The transition $t_0 \rightarrow t_1$ uses the *destination-month* driver $M(t_1)$, because the window rollover that defines $M(t_1)$ is what moves stocks between bins during that month. This makes the hindcast *conditional* on the realized market move — an exogenous-forcing run, not a lookahead-free forecast. A genuine forecast needs a forecast (or scenario) for $M(t_1)$.

2. Autocatalytic herding (the amplifier)

Mechanism. Winners attract flows; flows push prices; prices make winners. The herding rate should grow with the winner concentration itself — autocatalysis, not a constant rate.

Implementation.

$$h_{\text{eff}} = h_0 + h_1(w - l)$$

where w, l are the top/bottom momentum shares. $h_0 = 0$, $h_1 = 1.0$ **calibrated** (one-step hindcast error on pre-episode data: 2000-2019 for the 2020-21 case, 2000-2007 for the 2008-09 case). The signed form is deliberate: in a crash ($l \gg w$) it drives mass toward losers (panic selling); in a melt-up ($w \gg l$) toward winners (FOMO). It is symmetric by construction — the data decide the sign.

3. Sticky drift (the persistence kernel)

Mechanism. A concentrated distribution decays slower: when 98% of stocks are losers, the "average" transition matrix (estimated over all regimes) overstates the escape rate, because it mixes crash months with normal months.

Implementation.

$$T_{\text{eff}} = (1 - \lambda)T + \lambda I, \quad \lambda = \min[\max(w, l), 0.9]$$

$\lambda_1 = 1.0$ **calibrated** (same pre-episode protocol). When concentration is low, ordinary drift; when the distribution piles into an extreme bin, the drift freezes toward identity and the spike persists.

4. Spike-persistence damper (the decay-rate kernel) — new in rev. 3

Mechanism. Rev. 2 reproduced spike amplitudes but drained the extreme bins $\sim 2\times$ too fast after each peak. The physical reason: at a real peak the extreme bin's mass sits *deep* in the tail (Mar-2021: median top-bin momentum was 103%, only 11% within 8pp of the 60% boundary — measured). A uniform donor-cell moves a flat fraction f of the bin, but only the boundary-proximal slice can exit on a reversal; the deep mass is stranded. The outflow impedance should scale with how deep the mass is — i.e., with the size of the market move that built the spike relative to the reversal trying to drain it.

Implementation. Each extreme bin carries a *build-step memory* B (in pp): the characteristic $|M|$ that piled the mass there. B is reinforced by inflow ($|M|$ when M flows toward the bin) and decays toward neutral (one bin width, 0.15) with a 6-month time constant (assumed):

$$B \rightarrow 0.15 + (B - 0.15)e^{-1/6}, \quad B \rightarrow \max(B, |M|);$$

\text{on inflow}

When M *reverses* out of a spike (sign change) and the extreme source bin exceeds 50% share (gate, assumed — genuine spike state), its outflow fraction is damped:

$$\text{damp} = \min\left[\left(\frac{|M|}{B}\right)^p, 1\right], \quad p = 1.2$$

\text{(assumed, tuned for post-peak decay)}

A small reversal against a large build ($|M| \ll B$) barely dislodges the deep mass; a reversal comparable to the build drains efficiently. The 50% gate keeps the damper from firing during ordinary recoveries (e.g. mid-2020, when the loser bin held $\sim 30\%$): it engages only for true spikes — the 2021 top-bin melt-up and the 2008 bottom-bin crash.

Master equation

$$n(t+1) = T_{\text{eff}} n(t) + \text{advect}(n(t), M_{\text{cs}}, \text{damp}) + h_{\text{eff}} H(n(t)) + s - d n(t)$$

Drift (measured T , sticky), advection (measured M_{cs} , persistence- damped), herding (calibrated), entry/exit (measured).

Calibration (pre-episode only — no peeking)

episode	T window	h1, λ 1 fit on
2020-21	2000-2019 (measured)	one-step error, 2015-2019
2008-09	2000-2007 (measured)	one-step error, 2005-2007

$\kappa = 1.0$ is mechanical (a 15pp market move shifts momentum 15pp = one bin width); $w_{\text{eff}} = 0.15$ assumed. The persistence damper's $p = 1.2$, $\tau_{\text{build}} = 6$ mo, $\text{gate} = 0.5$ are assumed (tuned for post-peak decay, not fit on spike months). No parameter was fit on the spike months themselves.

Results — spikes reproduced at the right amplitude *and* decay

2020-21 (COVID crash + recovery)

- Peak: **actual 0.546 @ Mar-2021** → **predicted 0.515 @ Apr-2021** (amplitude error −6%, one month late).
- Post-peak decay: actual 0.406/0.326/0.357 @ +1/+2/+3 mo; predicted **0.418/0.275/0.239**. The +1-month decay is now exact (rev. 2: 0.259 — twice too fast); +2/+3 mo track the shape.
- Full-distribution RMSE: **0.0914** (rev. 2: 0.0920; rev. 1 best: 0.0821 — but rev. 1 never produced any spike: its peak was 0.09).

2020-21 hindcast

2008-09 (financial crisis)

- Peak: **actual 0.978 @ Mar-2009** → **predicted 0.994 @ Nov-2008** (amplitude error +1.6%; the actual is a plateau from late 2008 through Mar-2009 — the model captures its onset and height).
- Post-peak decay: actual 0.966/0.944/0.877; predicted **0.988/0.906/0.934**. +1 mo exact, +2/+3 mo within 4pp.
- Full-distribution RMSE: **0.1037** (rev. 2: 0.1182; rev. 1 best: 0.1110) — the best full-distribution score of any variant.

2008-09 hindcast

What changed vs rev. 2

	rev. 2	rev. 3 (persistence damper)
2021 +1-mo decay	0.259 (actual 0.406)	0.418
2021 +2-mo decay	0.173 (actual 0.326)	0.275
2008 +1-mo decay	0.983 (actual 0.966)	0.988
2008 +2-mo decay	0.605 (actual 0.944)	0.906
2008 RMSE	0.1182	0.1037

The damper fixes exactly what rev. 2 got wrong: the post-peak drain.

Mechanism in one line — deep spike mass cannot exit on a shallow reversal.

Limitations (stated plainly)

1. **Conditional, not lookahead-free.** The hindcast uses the realized $M(t_1)$. It answers "given the market moved M , where does the distribution go?" — the forcing is exogenous. A live forecast needs a market-move forecast alongside.
2. **2021 peak one month late.** The model puts the top-bin maximum in Apr-2021 vs the actual Mar-2021 — the mechanical translation lags the true (faster-than-monthly) repricing by one step.
3. **2008 maximum four months early.** The model peaks in Nov-2008; the actual maximum is Mar-2009 (a plateau). The model captures the plateau's onset, height, and decay but not its full duration.
4. **Damper parameters assumed.** $p = 1.2$, $\tau_{\text{build}} = 6$ mo, and the 50% gate are tuned for the two episodes' decay, not derived from first principles. A within-bin depth measurement (options-implied or high-frequency) would put the damper on measured footing.
5. **2021 +3-mo decay still fast.** Predicted 0.239 vs actual 0.357 — the damper releases as the bin drains below the gate; the actual shows a slower tail (likely idiosyncratic rotation the model lacks).

Data provenance

- Prices: Yahoo Finance daily closes, S&P 500 constituents, 2000-2021 (**measured**; survivorship bias noted — current constituents backfilled).
- Index additions/removals: entry/exit rates **measured** per episode.
- M_{cs} : cross-sectional window-delta **measured** from the same prices.
- T : monthly transition matrices **measured** per calibration window.
- $\kappa = 1.0$: mechanical identity (a pp is a pp). $w_{eff} = 0.15$ **assumed**.
- $h1 = 1.0$, $\lambda1 = 1.0$: **calibrated** on pre-episode one-step errors.
- Within-bin depth at peaks (Mar-2021 median 103%, 11% within 8pp): **measured** from the stock panel — motivates the damper.
- Damper $p = 1.2$, $\tau_{build} = 6$ mo, gate = 0.5: **assumed** (decay-tuned).
- Concentration-gated and fixed-tail outflow variants: tested in rev. 2, did not improve both episodes jointly — dropped.

Next steps

1. Measure within-bin depth directly (high-frequency or options data) to replace the assumed damper parameters.
2. Forecast $M(t+1)$ (VIX term structure? options-implied?) to make the hindcast a genuine forecast.
3. Test on a third episode (2000-02 dot-com, 2022 rate shock).
4. Push to GitHub (branch Stock-Market-Dynamics) once approved.

References

- Yahoo Finance (finance.yahoo.com) — daily OHLCV, S&P 500 constituents; used for momentum/volatility bins, transition matrices, the M_{cs} driver, and within-bin depth measurement.
- CBOE VIX, FRED series VIXCLS — regime labels in rev. 1 experiments.
- Project code: market_dynamics/src/model.py (master equation, four kernels), src/hindcast_final.py (frozen-config rev-3 runs).